

Hyper-Local Business Advisory and Financial Structuring Platform

Technical Design Document

1. Purpose and Scope

This document defines the technical design of an autonomous, agentic advisory platform for first-time rural and semi-urban entrepreneurs applying for concessional credit schemes. The platform combines a hyper-local business feasibility engine, a deterministic financial structuring and scheme-routing engine, and a post-disbursement monitoring and support layer, all driven through a single conversational interface built as an orchestrated multi-agent system.

The user interacts with one assistant and states an objective in natural language, in a regional language where required. The assistant plans and executes the required sequence of internal agents and tools, asks for missing information conversationally, and routes the user through profiling, feasibility assessment, financial structuring, funding preparation, and post-launch monitoring without requiring the user to navigate separate screens or modules manually.

2. Design Principles

- Single conversational entry point. All modules are reachable through one orchestrator agent rather than a fixed navigation structure.
- Deterministic core, reasoning periphery. All eligibility, project cost, loan amount, interest, tenure, and EMI calculations are computed by fixed rules and pure functions, never inferred by a language model. Language model agents interpret and explain these outputs but do not alter them.
- Open data first. Every data dependency is sourced from open government datasets, open-source geospatial infrastructure, or user-provided, consent-based personal data. No feature depends on a commercial data license or an institutional data-sharing agreement that would be unavailable to an independent team.
- Confidence-labeled outputs. Every agent output is tagged as either sourced from real data or estimated, so the user and any downstream reviewer can distinguish verified findings from inference.
- Rejection is a valid outcome. The system is permitted to advise against a business choice or flag it as not recommended, rather than always producing a positive recommendation.
- Persistent state across a long lifecycle. A single user profile and case state persists from initial profiling through post-disbursement monitoring, which may span months or years.

3. System Architecture Overview

The platform is structured as a directed graph of specialized agents and deterministic tools, coordinated by a central orchestrator, built on a graph-based agent orchestration framework. The graph is organized into three modules that together form one continuous session rather than three separate applications.

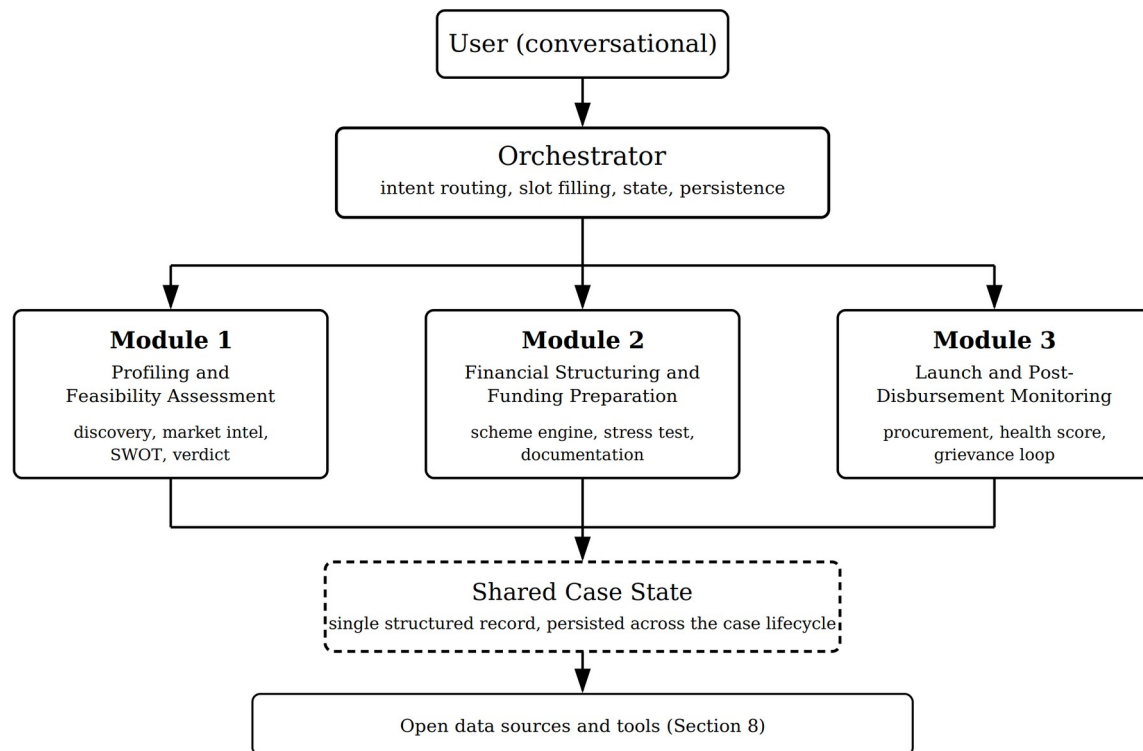


Figure 1. System architecture: orchestrator, the three modules, shared case state, and the underlying data layer.

3.1 Module 1: Profiling and Feasibility Assessment

Converts a conversational objective into a structured entrepreneur profile, identifies and ranks candidate business activities, produces a hyper-local market and risk assessment for the selected activity, and issues a feasibility verdict with an alternative recommendation path if the initial choice is not viable.

3.2 Module 2: Financial Structuring and Funding Preparation

Converts available margin capital into a project cost and loan eligibility figure, routes the case to the correct scheme tier, generates the repayment and moratorium schedule, stress-tests the plan against realistic downside scenarios, and prepares the documentation and application package required to apply.

3.3 Module 3: Launch and Post-Disbursement Monitoring

Supports the entrepreneur after funds are disbursed through procurement coordination, a launch roadmap, ongoing performance monitoring against the original feasibility model, and a grievance and outcome-tracking loop that improves future recommendations.

4. Orchestration Layer

The orchestrator is implemented as a stateful graph rather than a linear script, so that the assistant can move forward, branch, pause for missing information, and resume across sessions.

4.1 Core Responsibilities

- Intent classification: on every user turn, determine whether the user is continuing the current step, requesting a jump to a different part of the platform, or providing information the current step requires.
- Slot filling and clarification: identify missing required fields for the current step and ask for them conversationally, one at a time, rather than presenting a form.
- Agent and tool selection: choose which downstream agents to invoke based on the business category, data availability, and the stage of the case.
- State management: read and write a single structured case object shared across all agents, so no agent operates on stale or inconsistent information.
- Session persistence: checkpoint state so a case can be resumed after a gap of days or months, which is necessary because scheme disbursement and enterprise monitoring occur over a long timeline.

4.2 Control Flow Summary

The flow below describes the sequence of processing stages. It is not a fixed script; the orchestrator can re-enter any stage based on user intent.

- Entry and profiling: extract location, available margin capital, skills, assets, and business preference from conversational input into a structured profile.
- Business discovery: generate and rank candidate business activities against market, cost, and infrastructure constraints.
- Parallel local intelligence gathering: for the selected activity, run market reach, opportunity, risk, competitor, pricing, and supply chain analysis concurrently.
- Synthesis: merge all parallel outputs into a single structured assessment, including a strengths, weaknesses, opportunities, and threats synthesis.
- Adversarial review: run a dedicated review pass that can challenge the assessment and force reconsideration of the business choice.
- Feasibility verdict: classify the case as viable, marginal, or not recommended, with reasoning. If not recommended, return to business discovery with the rejected option excluded and a preference for adjacent, less saturated alternatives.

- Financial structuring: compute project cost, loan eligibility, scheme tier, and repayment schedule deterministically.
- Financial interpretation and stress testing: evaluate the deterministic output against seasonal revenue patterns and downside scenarios.
- Policy explanation: retrieve and explain the applicable scheme rules in plain language, without altering the deterministic eligibility decision.
- Documentation preparation: assist with registration and application form completion using information already collected.
- Post-disbursement monitoring: on re-entry after funding, track enterprise performance against the original plan and raise early warnings.
- Grievance and outcome logging: record issues raised and their resolutions, and feed outcomes back into future recommendation logic.

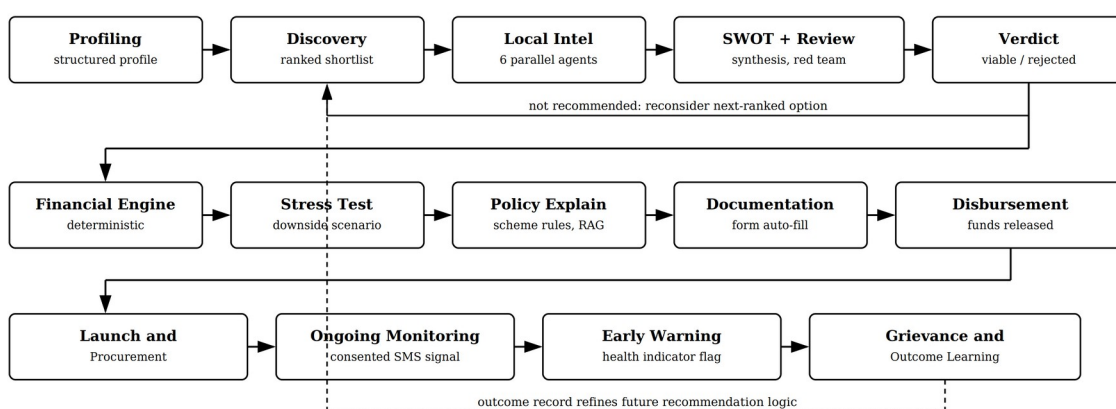


Figure 2. End-to-end control flow across all three modules, including the rejection loop back to business discovery and the outcome feedback loop from Module 3.

4.3 Language and Modality

All conversational agents operate through a multilingual natural language layer supporting text and voice input in regional languages in addition to English and Hindi. Translation and speech components are built on open-source Indic language models rather than a proprietary translation service, consistent with the open data and open tooling principle in Section 2.

5. Module 1 Detail: Profiling and Feasibility

5.1 Entrepreneur Profiling

Accepts free-form conversational input, in any supported language, describing available capital, skills, assets, land or premises, and business interest, and converts it into a structured profile with defined fields. Identifies constraints early, such as insufficient capital for a stated preference, so later stages do not evaluate an infeasible option.

5.2 Business Discovery and Ranking

Generates a shortlist of viable business activities for the profile by evaluating local market conditions, indicative setup and running costs, and locally available infrastructure. Produces a ranked list rather than a single suggestion, so that the user or the feasibility stage can move to the next-best option if the top choice fails review.

5.3 Hyper-Local Market Intelligence

For the selected business category, five analyses run concurrently and write to a shared case state:

- Market reach: resolves the stated village, block, or district to coordinates, disambiguating repeated place names using official location codes, then estimates the population and consumer base within a defined radius and identifies nearby markets and distribution points.
- Opportunity analysis: retrieves realistic sub-niches for the chosen sector from a curated reference collection of sector project reports, then cross-checks each niche against local competitor and demand data to classify saturation as low, medium, or high.
- Risk analysis: combines road and supply-route distance, seasonal price and demand variation over time, and a curated taxonomy of structural business risks such as single-buyer dependency into a plain-language set of risk flags.
- Competitor mapping: estimates the density of existing similar enterprises using a tiered fallback approach across multiple data sources, normalizes the result by local population, and benchmarks it against district and state averages.
- Pricing analysis: for categories with direct market price data, uses real recorded prices; for categories without direct data, estimates a representative local price range using a purchasing-power proxy, and explicitly labels which figures are real and which are estimated.

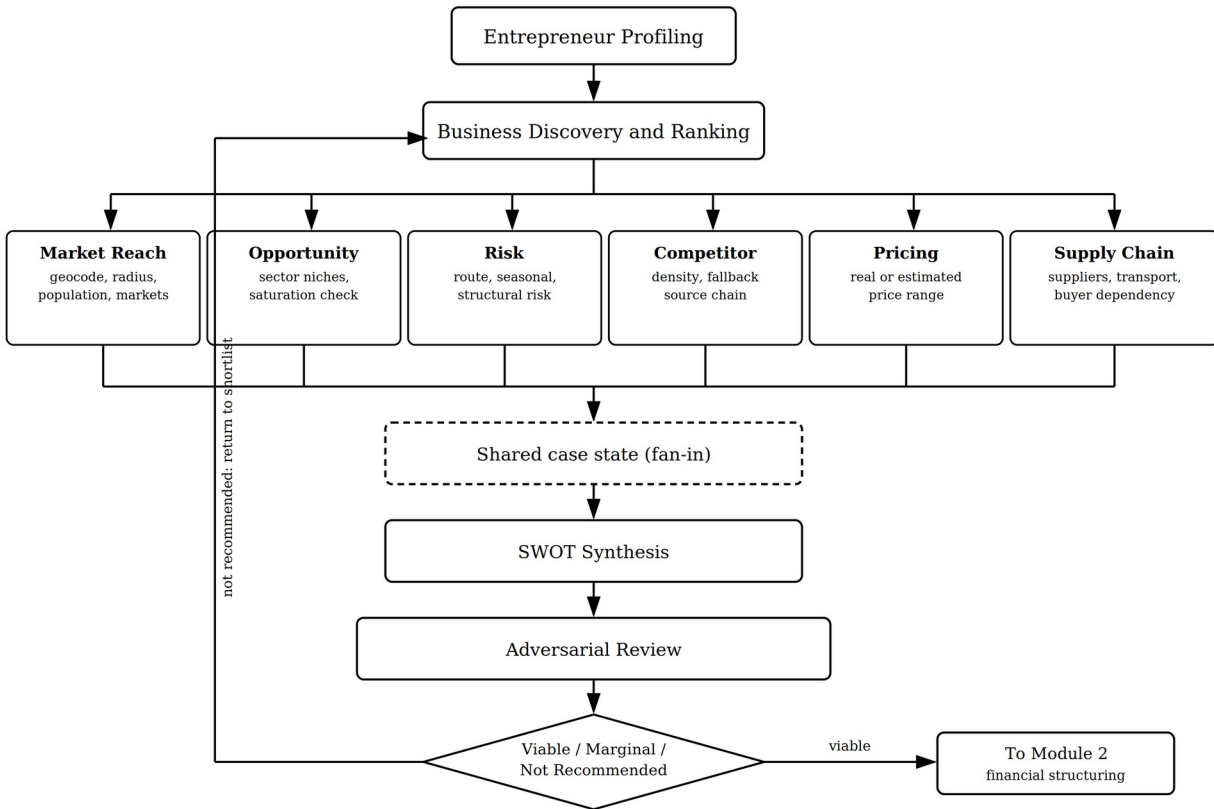


Figure 3. Module 1 internal flow: profiling and discovery, the six parallel local intelligence agents, fan-in synthesis, adversarial review, and the verdict branch.

5.4 Synthesis and Adversarial Review

Once all five analyses complete, a synthesis stage produces a structured strengths, weaknesses, opportunities, and threats assessment scaled to the specific budget, grounded against typical sector patterns rather than produced as unconstrained free text.

A separate adversarial review stage then examines the synthesis specifically to identify reasons the business may fail, and is authorized to reject the business choice outright rather than only listing weaknesses. This stage exists to counter the tendency, identified as the core problem in the underlying scheme, of beneficiaries choosing business activities based on anecdotal success rather than data.

5.5 Alternative Recommendation Path

If a business activity is rejected or classified as marginal, the system does not simply stop. It returns to the ranked shortlist and proposes the next viable alternative, with a preference for activities that are adjacent to the original interest but differ in the specific dimension that caused rejection, such as a less saturated niche within the same broad sector.

5.6 Supplementary Local Input

- The system records the entrepreneur's stated reason for choosing a business activity and factors this context into the feasibility narrative.
- The system supports structured feedback capture from previously funded entrepreneurs in the same business category and locality, which is incorporated as an additional evidence source in the opportunity and risk analyses.
- The system supports a lightweight local survey mechanism to capture on-ground observations from residents of the area, used to supplement gaps in official data sources.
- An interactive map view presents the geographic outputs of the market reach and competitor analyses for visual review.

6. Module 2 Detail: Financial Structuring and Funding Preparation

6.1 Deterministic Financial Engine

A fixed, non-inferential calculation component performs the following, based on the scheme structure described in the underlying problem statement:

- Computes total feasible project cost from stated available margin capital, treating the margin as a fixed contribution percentage of total project cost.
- Computes maximum loan eligibility as the remaining percentage of project cost.
- Applies fixed threshold logic to route the case to the applicable scheme tier based on computed project cost, applying the correct interest rate, repayment tenure, and moratorium period for that tier.
- Generates a period-by-period repayment schedule reflecting the applicable moratorium period before regular repayment begins.
- Estimates working capital requirements alongside the loan-funded capital expenditure.

6.2 Financial Interpretation

A reasoning stage separate from the deterministic engine evaluates whether the projected operating surplus for the business, informed by the pricing and market analyses from Module 1, is sufficient to comfortably service the computed repayment schedule. This stage explains the numbers in plain language and highlights margin of safety, but does not alter the deterministic figures.

6.3 Scenario and Stress Testing

The financial plan is tested against a downside scenario informed by the seasonal demand pattern identified in the risk analysis, to check whether repayment remains feasible during a low season rather than only under an average-case assumption.

6.4 Policy and Scheme Explanation

A retrieval-based reference stage explains the applicable scheme rules, required documentation, and process steps in plain language, sourced from the scheme guidelines and the underlying problem statement text. This stage explains rules; it does not determine eligibility, which remains the responsibility of the deterministic financial engine in Section 6.1.

6.5 Documentation and Application Preparation

A conversational, multilingual assistance stage helps the entrepreneur complete registration and loan application requirements, auto-filling fields already captured in the profile and financial plan, and tracks which required documents are complete, pending, or missing.

7. Module 3 Detail: Launch and Post-Disbursement Monitoring

7.1 Procurement Coordination

Identifies nearby entrepreneurs in the same business category, using the competitor density data already produced in Module 1, and coordinates pooled purchase requests for common raw materials or supplies to obtain better pricing than an individual entrepreneur could access alone.

7.2 Launch Support

Provides a structured launch roadmap covering supplier discovery, licensing steps, initial inventory planning, and early customer acquisition, derived from the feasibility and financial plans already produced.

7.3 Ongoing Performance Monitoring

With the entrepreneur's consent, the system reads incoming transaction notification messages on the entrepreneur's device to maintain an approximate, ongoing picture of business cash flow after launch. This data source is used for two purposes:

- Deriving an approximate creditworthiness indicator from income regularity and cash-flow consistency, for use alongside the deterministic scheme eligibility calculation, clearly labeled as an approximate indicator rather than a formal credit score.
- Comparing actual revenue, expense, and repayment patterns against the original feasibility and financial plan to compute an ongoing enterprise health indicator.

This data source is limited to transaction notification messages the entrepreneur's device receives and to which the entrepreneur has explicitly consented. It does not depend on access to bank systems, payment applications, or credit bureau records, and coverage is therefore approximate rather than complete.

7.4 Early Warning

When the ongoing enterprise health indicator falls outside an expected range relative to the original plan, the system raises a flag for review and suggests a specific type of intervention, such as a pricing adjustment, a supply chain change, or early outreach to a mentor.

7.5 Grievance and Outcome Learning

Entrepreneurs can raise issues through the same conversational interface, which are logged with the associated case context. The system tracks whether a given intervention was followed by an improvement in the enterprise health indicator, and this outcome record is used to refine future feasibility and recommendation logic for similar profiles.

Because this feedback loop has no historical record at initial deployment, it is seeded with a synthetic outcome dataset constructed to represent plausible intervention and outcome patterns. This synthetic dataset is clearly labeled as such throughout the system and is progressively replaced by genuine outcome records as real cases accumulate.

8. Data Sources

All data sources below are open government datasets, open-source geospatial infrastructure, open-source language models, or data provided directly by the user with consent. No feature depends on a commercial data license, a credit bureau, or an institutional data-sharing agreement.

Category	Source	Used for
Geocoding	OpenStreetMap Nominatim	Resolving village, block, or district to coordinates
Location codes	Local Government Directory	Disambiguating repeated place names across states
Administrative boundaries	Open government geospatial datasets and community-maintained boundary repositories	District and block boundary geometry
Population	Census village-level datasets	Estimating local consumer base
Purchasing power proxy	Socio-Economic Caste Census asset data	Estimating regional purchasing power
Points of interest	OpenStreetMap Overpass API	Nearby markets, shops, and infrastructure
Registered enterprises	Udyam Registration open dataset	Competitor density estimation
Commodity prices	Government commodity price open data API	Real agricultural pricing and seasonal patterns
Road distance and routing	Open-source routing engine over open map data	Supply chain and accessibility risk
Sector reference material	Publicly published sector and model project reports	Business sub-niche identification
Industry classification	National industrial classification reference tables	Standardizing business category taxonomy
Scheme rules	Published scheme guidelines and the governing problem statement	Eligibility logic and plain-language explanation
Regional language processing	Open-source Indic language translation and speech models	Multilingual conversational interface
Transaction notifications	User device, with explicit consent	Approximate cash-flow monitoring after launch
Outcome history	Synthetic seed dataset, replaced progressively by real records	Cold start for the grievance and outcome learning loop

9. State and Data Model

A single structured case object is shared across all agents and persists for the lifetime of the case. It is organized into the following sections, each owned and written by the corresponding module described above:

- Entrepreneur profile: location, available capital, skills, assets, and stated business preference and reason.
- Business shortlist: ranked candidate activities with supporting rationale.
- Market intelligence: the combined output of the market reach, opportunity, risk, competitor, and pricing analyses, each carrying a source confidence label.
- Feasibility record: the synthesis, adversarial review outcome, and final verdict, including a history of any rejected options and the reasoning for rejection.
- Financial plan: project cost, loan eligibility, scheme tier, repayment schedule, and stress test results.
- Application status: document checklist and completion state.
- Monitoring record: a time series of enterprise health indicators from post-disbursement tracking.
- Grievance and outcome log: issues raised, interventions suggested, and observed outcomes.

10. Known Constraints and Assumptions

- All financial eligibility and repayment figures are produced by fixed rules and are not subject to language model interpretation or adjustment.
- Approximate indicators derived from transaction notification messages are explicitly labeled as approximate and are not presented as a formal credit score or a substitute for a credit bureau record.
- Coverage of the transaction monitoring feature is limited to notification messages received on the entrepreneur's device and depends on the entrepreneur's continued consent.
- Sector reference material, purchasing power proxies, and price estimates for categories without direct market data are labeled as estimates and distinguished from figures sourced from real, current data.
- The outcome learning loop operates on a synthetic seed dataset until sufficient genuine case outcomes accumulate, and this status is visible wherever outcome-informed recommendations are shown.
- Rural point-of-interest and competitor data is sparse by nature, and the competitor mapping and market reach analyses rely on a tiered fallback across multiple open sources rather than a single authoritative dataset.

